

Course Syllabus

CS 270 Mathematical Foundations in CS - Syllabus

Term and Credits

Winter 2025-2026

3 Credits

Room and Time

Section	Days	Time	Room	Instructor	Course Assistant
001	TR	12:00 pm - 01:20 pm	3675 Market Street Room 1103	Mark Boady	Shehryar Usman
002	TR	03:00 pm - 04:20 pm	3675 Market Street Room 1104	Mark Boady	Shehryar Usman

Instructors

Professor Mark Boady

Electronic Mail Address: mwb33@drexel.edu (<mailto:mwb33@drexel.edu>)

Office: 3675 Market Street Room 1063 (near snack machine)

Office Hours: 2-3PM Tuesday and Thursday (or by appointment)

Teaching Assistant(s)

All TA Office Hours are held in the CLC <https://www.cs.drexel.edu/clc>  (<https://www.cs.drexel.edu/clc>)

Name	Email	Hours
Shehryar Usman	su83@drexel.edu (mailto:su83@drexel.edu)	Monday 4-6PM and Wednesday 4-6PM in CLC (3675 Market Street 1066)

Course Description

Introduces formal logic and its connections to Computer Science. Students learn to translate statements about the behavior of computer programs into logical claims and to prove such assertions using both traditional techniques and automated tools. Considers approaches to proving termination, correctness, and safety for programs. Discusses propositional and predicate logic, logical inference, recursion and recursively defined sets, mathematical induction, and structural induction.

Course Objective and Goals

1. To use recursion and divide and conquer to solve problems
2. To provide recursive definitions of patterns and data structures
3. To formally specify the input/output requirements of programs
4. To use induction and other proof techniques to prove properties of algorithms, data structures, programs, and computer systems
5. To use logic to describe the state of systems and to use logical deduction (by hand and using tools) to prove properties of systems
6. To understand the power and limitations of formal logic.

Topics

1. Functional Programming
2. Recursion, Recursive Definitions and Induction
3. Propositional and Predicate Logic
4. Formal Proof using Natural Deduction
5. Applications of Logic to Computer Science
6. Divide and Conquer Algorithms and Recurrence Relations
7. Program Specification and Verification
8. Automated Reasoning
9. Test Case and Counter Example Generation

Audience and Purpose within Plan of Study

This is a required course for all Computer Science and Software Engineering students. It should also be of interest to Computer Engineering, Mathematics students and students with an interest in logic and computation.

Prerequisites

CS 172 Minimum Grade: D or CS 176 Minimum Grade: D or CS 265 Minimum Grade: D or SE 103 Minimum Grade: D or ECEC 301 Minimum Grade: D or ECEC 201 Minimum Grade: D

What Students Should Know Prior to this Course

1. Ability to read and understand code.
2. Basic understanding of program execution.
3. Ability to write simple recursive programs.

What Students will be able to do upon Successfully Completing this Course:

1. Use Proofs by Deduction to Justify Logical Statements
2. Be able to write and analyze Recursive Functions
3. Be able to implement and use a SAT solver.
4. Use Inductive Proofs to Justify the correctness of programs and statements.
5. Use logic to describe the state of systems.

Textbook

Professor Boady's Readings: <https://logic.boady.net/README.html>  (<https://logic.boady.net/README.html>)

Optional Resources:

Book of Proof (Second Edition)

Richard Hammack

Paperback: ISBN 978-0-9894721-0-4

Hardcover: ISBN 978-0-9894721-1-1

Available for Free online at: <http://www.people.vcu.edu/~rhammack/BookOfProof/> 
(<http://www.people.vcu.edu/~rhammack/BookOfProof/>)

The Racket Guide

Matthew Flatt, Robert Bruce Findler and PLT

<https://docs.racket-lang.org/guide/index.html>  [\(https://docs.racket-lang.org/guide/index.html\)](https://docs.racket-lang.org/guide/index.html)

Forall x: Calgary

P.D. Magnus and Tim Button

<http://forallx.openlogicproject.org>  [\(http://forallx.openlogicproject.org\)](http://forallx.openlogicproject.org)

If you want to learn more about functional programming.

The Little Schemer - 4th Edition

Daniel P. Friedman and Matthias Felleisen

ISBN-13: 978-0262560993

ISBN-10: 0262560992

Available at: [Amazon](https://www.amazon.com/Little-Schemer-Daniel-P-Friedman/dp/0262560992/ref=sr_1_1?keywords=little+scheme&qid=1568217153&s=gateway&sr=8-1)  [\(https://www.amazon.com/Little-Schemer-Daniel-P-Friedman/dp/0262560992/ref=sr_1_1?keywords=little+scheme&qid=1568217153&s=gateway&sr=8-1\)](https://www.amazon.com/Little-Schemer-Daniel-P-Friedman/dp/0262560992/ref=sr_1_1?keywords=little+scheme&qid=1568217153&s=gateway&sr=8-1)

If you want to learn more about logic and proofs.

Connecting Discrete Mathematics and Computer Science - 2nd Edition

David Liben-Nowell

ISBN-13: 978-1009150491

Available at: [Amazon](https://www.amazon.com/Connecting-Discrete-Mathematics-Computer-Science-ebook/dp/B09VXPWYJ5?ref=ast_author_mpb)  [\(https://www.amazon.com/Connecting-Discrete-Mathematics-Computer-Science-ebook/dp/B09VXPWYJ5?ref=ast_author_mpb\)](https://www.amazon.com/Connecting-Discrete-Mathematics-Computer-Science-ebook/dp/B09VXPWYJ5?ref=ast_author_mpb)

Grading and Policies

- Homeworks: 30%
- Labs: 20%
- Midterm Exam: 25%
- Final Exam: 25%

Final grades will be determined by your total points weighted according to this distribution. Grades may be curved but are generally computed via the formula below. It may be modified at the instructor's sole discretion, but letter grades will generally not be lower than those shown here.

- [100-97] A+
- (97-93] A
- (93-90] A-
- (90-87] B+
- (87-83] B
- (83-80] B-
- [80-77] C+
- (77-73] C
- (73-70] C-
- (70-67] D+
- (67-60] D
- (60-0] F

Late Policy

- Midterm/Final Exam
 - Requires approval from the Professor to take late.
- Labs
 - Requires approval from the Professor to complete late.
 - Must be completed within 7 days of the missed class.
 - Must be completed in person during Professor or Course Assistant office hours.
- Homeworks
 - Penalties:
 - -10% 1 day late
 - -20% 2 days late
 - -30% 3 days late
 - -40% 4 days late
 - Not Accepted after 4 days
- **Late Passes:** Each student has two (2) late passes. You may exchange a **late pass** for 1 extra day on a **homework** for any reason. You will receive one extra day with no penalty. You **must** inform a Professor before the deadline for the assignment you are

using the late pass on by using the "Late Pass Request Form" linked on the course webpage. You **may** use both late passes on the same assignment to get 2 extra days.

- **Special Exceptions:** In exceptional circumstances, the **Professor** may wave additional penalties or accept work past the cut-off. Please message the Professor directly to request a special exception.

Academic Honesty Policy

The CCI Academic Honesty policy is in effect for this course. Please see the policy at <http://drexel.edu/cci/resources/current-students/undergraduate/policies/cs-academic-integrity/>  (<http://drexel.edu/cci/resources/current-students/undergraduate/policies/cs-academic-integrity/>).

Academic Honesty Violations **will** be reported to the University. Punishment will be determined by the severity of the incident. Punishments include, but are not limited to,

- Failing grade for class
- Deduction of one letter grade
- Zero on Assignment/Exam Violation took place on

Course Material

Programming Language

- This class will use the Racket Programming Language.
- This class will use the DrRacket IDE for development.
- You may use Racket from the command line instead of the IDE.
- [Download Racket](http://racket-lang.org)  (<http://racket-lang.org>)

Lectures

- Each class will start with a lecture.
- Most classes will have a short lab following the lecture.

Labs

- You may work in groups of up to 2 students on labs.
- Labs are intended to help you learn the material.
- Collaboration with other students during labs is encouraged.
- Lab sheets are due at the end of the class period.
- If you miss a class, contact your Professor to make up the lab.
- The lowest lab grade for the term will be dropped.

Homeworks

- Most Weeks will have a homework assignment.
- See Schedule for Due Dates
- All homeworks are submitted to Gradescope
- Although help may be sought from the instructor/TA, homework is intended to be done individually without assistance from any other external sources

Exams

- There will be two Exams.
- Both will be held in class.
- Midterm: Week 6
- Final: Finals Week

Extra Credit

- In this course, we value the course survey results. It is important for you to receive grades so you can determine how you are performing in class. We also want to know how we are doing. The course survey is our most important tool for determining how the students feel about the class. We also understand that completing the survey takes time and effort. Due to the importance of the course survey, extra credit will be provided to those students who complete it and provide evidence it has been completed. The extra credit will be 1 point added to the final grade for the class. This will take place at the end of the term.

PERMITTED USE OF ARTIFICIAL INTELLIGENCE TOOLS IN THIS COURSE: Artificial intelligence tools such as large language models (e.g., ChatGPT) are not permitted to be used in preparing submitted work for this course. Further information on university

policies regarding the use of artificial intelligence tools in the classroom is available in the Draft Drexel University Policy on Academic Integrity Pertaining to Artificial Intelligence. Artificial intelligence tools may be used as study aids. They may be used to help prepare for tests, study material, or learn about topics. They may not be used to generate any code/answers that are submitted for credit.

General Information

University Policies

In addition to the course policies listed on this syllabus, course assignments or course website, the following University policies are in effect:

- Computer/Software Help provided by iCommons: <https://drexel.edu/cci/current-students/icommons/> ➞ [\(https://drexel.edu/cci/current-students/icommons/\)](https://drexel.edu/cci/current-students/icommons/)
- Academic Integrity: <https://drexel.edu/studentlife/community-standards/code-of-conduct/academic-integrity-policy> ➞ [\(https://drexel.edu/studentlife/community-standards/code-of-conduct/academic-integrity-policy\)](https://drexel.edu/studentlife/community-standards/code-of-conduct/academic-integrity-policy)
- Academic Integrity Pertaining to Artificial Intelligence: https://drexel.edu/provost/policies-calendars/policies/academic_integrity_artificial_intelligence/ ➞ [\(https://drexel.edu/provost/policies-calendars/policies/academic_integrity_artificial_intelligence/\)](https://drexel.edu/provost/policies-calendars/policies/academic_integrity_artificial_intelligence/)
- Academic Integrity Conduct Process: <https://drexel.edu/studentlife/community-standards/code-of-conduct/academic-integrity-policy/conduct-process> ➞ [\(https://drexel.edu/studentlife/community-standards/code-of-conduct/academic-integrity-policy/conduct-process\)](https://drexel.edu/studentlife/community-standards/code-of-conduct/academic-integrity-policy/conduct-process)
- Official Final Exam Schedule: <http://www.drexel.edu/registrar/scheduling/exams/> ➞ [\(http://www.drexel.edu/registrar/scheduling/exams/\)](http://www.drexel.edu/registrar/scheduling/exams/)
- Students with Disability Statement: <https://drexel.edu/disability-resources/support-accommodations/student-family-resources> ➞ [\(https://drexel.edu/disability-resources/support-accommodations/student-family-resources\)](https://drexel.edu/disability-resources/support-accommodations/student-family-resources)
- Course Drop Policy: <https://drexel.edu/provost/policies-calendars/policies/course-add-drop/> ➞ [\(https://drexel.edu/provost/policies-calendars/policies/course-add-drop/\)](https://drexel.edu/provost/policies-calendars/policies/course-add-drop/)
- Drexel Student Learning Priorities: <https://drexel.edu/institutionalresearch/assessment/outcomes/dslp/> ➞ [\(https://drexel.edu/institutionalresearch/assessment/outcomes/dslp/\)](https://drexel.edu/institutionalresearch/assessment/outcomes/dslp/)
- The instructor may, at his/her/their discretion, change any part of the course during the term, including assignments, grade breakdowns, due-dates, and the schedule. Such changes will be communicated to students via the course web site Announcements

page. This page should be checked regularly and frequently for such changes and announcements. Other announcements, although rare, may include class cancellations and other urgent announcements.

CCI Culture and Community Statement

CCI's Commitment to Culture and Community The College of Computing and Informatics commits to creating a positive and safe learning environment for everyone – instructors, professional staff, and students – both inside and outside of the classroom. We embrace the wide range of thoughts, perspectives, and experiences that each community member brings, and we honor everyone's unique identity and lived experience. We encourage each community member to share information and feedback that will assist instructors and staff in fostering a supportive and inclusive community environment. For more information about CCI's commitment to culture and community, visit [Culture & Community Council | Drexel CCI](https://drexel.edu/cci/about/culture-community-council/)  [\(https://drexel.edu/cci/about/culture-community-council/\)](https://drexel.edu/cci/about/culture-community-council/)

Appropriate Use of Course Materials

It is important to recognize that some or all of the course materials provided to you are the intellectual property of Drexel University, the course instructor, or others. Use of this intellectual property is governed by Drexel University policies, including the IT-1 policy found at: <https://drexel.edu/it/about/policies/policies/01-Acceptable-Use/>  [\(https://drexel.edu/it/about/policies/policies/01-Acceptable-Use/\)](https://drexel.edu/it/about/policies/policies/01-Acceptable-Use/) Briefly, this policy states that all course materials including recordings provided by the given prior written approval by the University. Doing so may be considered a breach of this policy and will be investigated and addressed as possible academic dishonesty, among other potential violations. Improper use of such materials may also constitute a violation of the University's Code of Conduct found at: <https://drexel.edu/studentlife/community-standards/code-of-conduct>  [\(https://drexel.edu/studentlife/community-standards/code-of-conduct\)](https://drexel.edu/studentlife/community-standards/code-of-conduct) and will be investigated as such.

Recording of Class Activities:

In general, students and others should not record course interactions and course activities in lecture, lab, studio or recitation. Students who have an approved accommodation from the Office of Disability Resources to record online lectures and discussions for note taking purposes should inform their course instructor(s) of their approved accommodation in advance. The recording of lectures and discussions may only be carried out by the students enrolled in the class who have an approved accommodation from Disability Resources with their instructors' prior knowledge and consent. Students with approved accommodations may be asked to turn off their recorder if confidential or personal information is presented.

If a student has any comments, concerns, or questions about provided class materials and/ or recording, talk to your course instructor first. If this does not resolve the issue, you can also reach out to the Department Head, and use the process described for a grade

appeal to move your concern forward. The process described for grade appeals can be found at: <https://drexel.edu/provost/policies-calendars/policies/grade-appeals/>  (<https://drexel.edu/provost/policies-calendars/policies/grade-appeals/>)

Tentative Course Schedule

Please see the appropriate assignment webpages for a detailed description of course deliverables.

Week	Topic	Assignments
(1) 1/05/2026	Lecture 1: Formal Logic Lecture 2: Natural Deduction Basics	Lab 1 - Due at end of Lecture 1 Lab 2 - Due at end of Lecture 2 Homework 1 - Due 11:59PM Friday
(2) 1/12/2026	Lecture 3: Disjunction Elimination Lecture 4: Proofs by Contradiction	Lab 3 - Due at end of Lecture 3 Lab 4 - Due at end of Lecture 4 Homework 2 - Due 11:59PM Friday
(3) 1/19/2026	Lecture 5: Predicates and First Order Logic Lecture 6: Deduction with Predicates <i>Note: Univ. Holiday Monday</i>	Lab 5 - Due at end of Lecture 4 Lab 6 - Due at end of Lecture 5 Homework 3 - Due 11:59PM Friday
(4) 1/26/2025	Lecture 7: Introduction to Racket Lecture 8: Recursion and Lists	Lab 7 - Due at end of Lecture 7 Lab 8 - Due at end of Lecture 8 Homework 4 - Due 11:59PM Friday
(5) 2/2/2026	Lecture 9: Equational Reasoning Lecture 10: High Order Functions	Lab 9 - Due at end of Lecture 9 Lab 10 - Due at end of Lecture 10 Homework 5 - Due 11:59PM Friday
(6) 2/09/2026	Lecture 11: Midterm Review Midterm	Lab 11 - Due at end of Lecture 11 Midterm - In Class During Second Lecture of Week

Week	Topic	Assignments
(7) 2/16/2026	Lecture 12: Mathematical Induction Lecture 13: Peano Arithmetic	Lab 12 - Due at end of Lecture 12 Lab 13 - Due at end of Lecture 13 Homework 6 - Due 11:59PM Friday
(8) 2/23/2026	Lecture 14: Induction on Lists Lecture 15: Lists as Trees	Lab 14 - Due at end of Lecture 14 Lab 15 - Due at end of Lecture 15 Homework 7 - Due 11:59PM Friday
(9) 3/2/2026	Lecture 16: Boolean Normal Forms Lecture 17: Expressions as Trees	Lab 16 - Due at end of Lecture 16 Lab 17 - Due at end of Lecture 17 Homework 8 - Due 11:59PM Friday
(10) 3/9/2026	Lecture 18: SAT Solvers Lecture 19: NP-Complete Problems	Lab 18 - Due at end of Lecture 18 Lab 19 - Due at end of Lecture 19 Homework 9 - Due 11:59PM Friday Extra Credit Course Survey - Due 11:59PM Friday
(Finals) 3/16/2026	Final Exam FRIDAY March 20, 2026 from 3:30-5:30PM in 1054-1055 of 3675 Market Street	